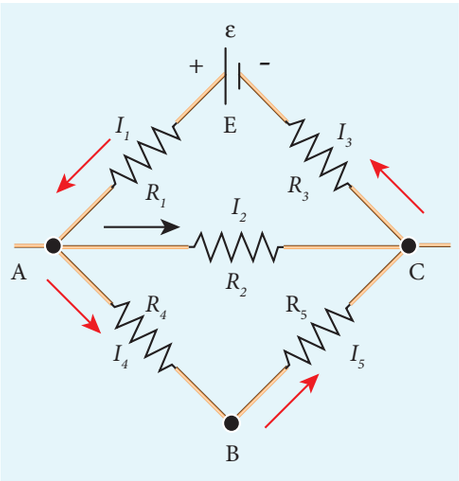
Kirchhoff voltage rule has to be applied only when all currents in the circuit reach a steady state condition (the current in various branches are constant).

EXAMPLE 2.21

The following figure shows a complex network of conductors which can be divided into two closed loops like EACE and ABCA. Apply Kirchhoff's voltage rule (KVR),



Solution

Thus applying Kirchhoff's second law to the closed loop EACE

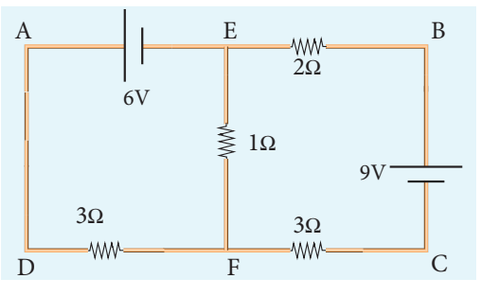
$$I_1R_1 + I_2R_2 + I_3R_3 = \varepsilon$$

and for the closed loop ABCA

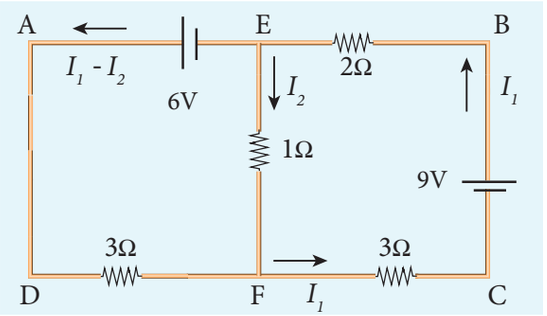
$$I_4R_4 + I_5R_5 - I_2R_2 = 0$$

EXAMPLE 2.22

Calculate the current that flows in the 1 Ω resistor in the following circuit.



Solution



We can denote the current that flows from 9V battery as I_1 and it splits up into I_2 and $(I_1 - I_2)$ at the junction E according Kirchhoff's current rule (KCR).

Now consider the loop EFCBE and apply KVR, we get

$$1I_2 + 3I_1 + 2I_1 = 9$$

$$5I_1 + I_2 = 9 \qquad (1)$$

Applying KVR to the loop EADFE, we get

$$3(I_1 - I_2) - 1I_2 = 6$$

$$3I_1 - 4I_2 = 6 \qquad (2)$$

Solving equation (1) and (2), we get

$$I_1 = 1.83 \text{ A and } I_2 = -0.13 \text{ A}$$

It implies that the current in the 1 ohm resistor flows from F to E.

2.5.3 Wheatstone's bridge

An important application of Kirchhoff's rules is the Wheatstone's bridge. It is used to compare resistances and in determining the unknown resistance in electrical network. The bridge consists of four resistances *P, Q, R* and *S* connected as shown in Figure 2.25. *A* galvanometer *G* is connected between the points *B* and *D*. The battery is connected between the points *A* and *C*. The current

12th Physics_Vol-1_Unit 2_English Medium.indd 106 12-12-2022 17:24:48